

# SIMATS ENGINEERING

## ASSESSMENT TOOL 3

**COURSE NAME:** CSA16 **COURSE CODE:** Data Warehousing and Data Mining

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### COURSE OUTCOME COVERED

**CO4:** Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

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**QUESTIONS: 05**

**Total Marks:50**

Annexure A

Industry Problem Solving Task

| <i>Criteria</i>       | Understandin<br>g of Industry<br>Problem (2) | Application<br>of<br>Engineering<br>Concepts<br>(2.5) | Solutio<br>n<br>Design<br>(2.5) | Use of<br>Tools (2) | Analysis (1) | <i>Total(10<br/>)</i> |
|-----------------------|----------------------------------------------|-------------------------------------------------------|---------------------------------|---------------------|--------------|-----------------------|
| <i>Weightag<br/>e</i> | 20%                                          | 25%                                                   | 25%                             | 20%                 | 10%          | 100%                  |
| <i>Q1</i>             |                                              |                                                       |                                 |                     |              |                       |
| <i>Q2</i>             |                                              |                                                       |                                 |                     |              |                       |
| <i>Q3</i>             |                                              |                                                       |                                 |                     |              |                       |
| <i>Q4</i>             |                                              |                                                       |                                 |                     |              |                       |
| <i>Q5</i>             |                                              |                                                       |                                 |                     |              |                       |

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### ***Code of Conduct:***

*I M MEGHANADH(192325026) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***  
**MEGHANADH**

## RUBRICS FOR CALCULATION:

| Criteria                            | Weightage | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                               |
|-------------------------------------|-----------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|---------------------------------------------------|
| Criteria                            | Weightage | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                               |
| Understanding of Industry Problem   | 20%       | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |
| Application of Engineering Concepts | 25%       | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts           |
| Solution Design                     | 25%       | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                   |
| Use of Tools                        | 15%       | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                 |
| Analysis                            | 10%       | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                               |
| Professionalism                     | 5%        | Highly professional, well-documented, and clearly presented work             | Good documentation and presentation                  | Basic documentation with some gaps             | Poor documentation and presentation               |

## Annexure A

Q1 . Dataset: Supermarket Purchases

| Customer | Visits/Month | Avg Spend (₹) | Items Bought |
|----------|--------------|---------------|--------------|
| C1       | 10           | 2000          | 15           |
| C2       | 4            | 5000          | 8            |
| C3       | 12           | 1800          | 18           |
| C4       | 3            | 6000          | 6            |
| C5       | 11           | 2100          | 16           |

### Question:

Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. (BL3 – 7 Marks)

Q2. Dataset: Credit Card Transactions

| Transaction | Amount (₹) | Location | Time (hrs) |
|-------------|------------|----------|------------|
| T1          | 200        | Chennai  | 10         |
| T2          | 5000       | Delhi    | 2          |
| T3          | 150        | Chennai  | 11         |
| T4          | 7000       | Mumbai   | 1          |
| T5          | 180        | Chennai  | 12         |

### Question:

Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. (BL4 – 7 Marks)

Q3. Dataset: Telecom Usage

| User | Call Duration (min) | Data Usage (GB) | Recharge (₹) |
|------|---------------------|-----------------|--------------|
| U1   | 300                 | 5               | 200          |
| U2   | 100                 | 2               | 100          |
| U3   | 350                 | 6               | 220          |
| U4   | 80                  | 1               | 90           |
| U5   | 320                 | 5.5             | 210          |

### Question:

Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. (BL4 – 7 Marks)

Q4 . Dataset: E-commerce Behavior

**User Product Views Clicks Purchases**

|    |    |    |   |
|----|----|----|---|
| U1 | 50 | 20 | 5 |
| U2 | 30 | 10 | 2 |
| U3 | 60 | 25 | 6 |
| U4 | 20 | 8  | 1 |
| U5 | 55 | 22 | 5 |

**Question:**

Suggest a clustering technique to group users and explain how it improves recommendation systems. (BL3 – 7 Marks)

Q5. Dataset: Traffic Data

**Location Vehicle Count Avg Speed (km/h) Delay (min)**

|    |     |    |    |
|----|-----|----|----|
| L1 | 200 | 30 | 10 |
| L2 | 500 | 15 | 25 |
| L3 | 220 | 28 | 12 |
| L4 | 600 | 10 | 30 |
| L5 | 210 | 32 | 9  |

**Question:**

Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. (BL4 – 7 Marks)

**Q1. Supermarket Purchases – Customer Segmentation**

A suitable approach is **K-Means clustering** using the three numerical features:

**Visits/Month, Average Spend, and Items Bought.**

**Step 1: Analyze the data**

| Customer | Visits | Avg Spend (₹) | Items | Customer Type             |
|----------|--------|---------------|-------|---------------------------|
| C1       | 10     | 2,000         | 15    | Frequent, low-spending    |
| C2       | 4      | 5,000         | 8     | Infrequent, high-spending |
| C3       | 12     | 1,800         | 18    | Frequent, low-spending    |
| C4       | 3      | 6,000         | 6     | Infrequent, high-spending |
| C5       | 11     | 2,100         | 16    | Frequent, low-spending    |

**Step 2: Preprocess**

Since average spending has a much larger numerical scale than visits and items, **normalize/standardize the features** before applying K-Means.

**Step 3: Select clusters**

A suitable choice is **K = 2**:

- **Cluster 1:** C1, C3, C5 → frequent visitors, lower average spending, and more items purchased.
- **Cluster 2:** C2, C4 → fewer visits, but higher average spending and fewer items.

**Step 4: Targeted Marketing**

**Cluster 1 – Frequent shoppers**

- Offer loyalty points.
- Provide frequent-purchase discounts.
- Recommend complementary products.
- Encourage larger basket sizes.

**Cluster 2 – High-value occasional shoppers**

- Provide premium offers.
- Send personalized high-value product recommendations.
- Offer incentives to increase visit frequency.
- Use loyalty programs to encourage repeat purchases.

### Conclusion

K-Means with **two customer segments** helps the supermarket understand purchasing patterns and create **personalized marketing campaigns**, improving customer engagement, sales, and retention.

## Q2. Credit Card Transactions – Fraud Detection

A clustering-based fraud detection system can identify transactions that are significantly different from normal transaction patterns.

### Step 1: Select Features

Use:

- Transaction Amount
- Location
- Transaction Time

Since **Location is categorical**, it should be encoded appropriately before clustering.

### Step 2: Normalize Data

Transaction amount and time have different scales, so numerical features should be **standardized** before calculating distances.

### Step 3: Apply Clustering

A suitable method is **DBSCAN**, because it can identify dense groups of normal transactions and mark isolated observations as **noise/outliers**.

From the dataset:

- **T1, T3, T5:** Chennai, low transaction amounts, daytime transactions → similar pattern.
- **T2:** ₹5,000, Delhi, 2 hrs → unusual compared with the Chennai transactions.
- **T4:** ₹7,000, Mumbai, 1 hr → also significantly different.

Therefore, T2 and T4 can be flagged as **potentially unusual transactions**.

### Step 4: Fraud Detection Process

**Transaction Data → Preprocessing → Encoding & Normalization → DBSCAN → Normal Clusters + Outliers → Fraud Investigation**

### Conclusion

Clustering can identify unusual transactions without requiring a previously labeled fraud dataset. **DBSCAN is suitable because it detects dense normal patterns and isolated outliers**, which can then be sent for additional fraud verification.

## Q3. Telecom Usage – Churn-Prone Users

Clustering can group telecom users according to their **Call Duration, Data Usage, and Recharge Amount**.

### Step 1: Apply K-Means

First, normalize the three numerical features and apply **K-Means clustering**.

The data shows two clear behavioral groups:

- **High-usage users:** U1, U3, U5
- **Low-usage/low-recharge users:** U2, U4

| User | Calls | Data | Recharge | Segment    |
|------|-------|------|----------|------------|
| U1   | 300   | 5    | ₹200     | High usage |
| U2   | 100   | 2    | ₹100     | Low usage  |
| U3   | 350   | 6    | ₹220     | High usage |
| U4   | 80    | 1    | ₹90      | Low usage  |
| U5   | 320   | 5.5  | ₹210     | High usage |

### Step 2: Identify Churn-Prone Group

Users with **low usage and low recharge** can be considered potentially churn-prone because

their engagement with the telecom service is relatively low.  
Thus, **U2 and U4** can be placed in a higher-priority retention segment.

### Step 3: Retention Strategies

For low-usage users:

- Offer affordable recharge plans.
- Provide additional data or call benefits.
- Send personalized offers.
- Provide loyalty rewards.
- Monitor changes in usage patterns.

For high-usage users:

- Offer premium plans.
- Provide higher data limits.
- Reward continued usage.

### Conclusion

Clustering helps the telecom company identify different usage segments and focus retention efforts on **low-engagement users such as U2 and U4**. This enables personalized offers and can help reduce customer churn.

## Q4. E-Commerce Behavior – Recommendation System

A suitable technique is **K-Means clustering**, because the dataset contains numerical behavioral features: **Product Views, Clicks, and Purchases**.

### Step 1: Normalize the Data

The three features should be standardized so that no feature dominates the distance calculation.

### Step 2: Form Customer Groups

The data suggests:

- **Highly engaged users:** U1, U3, U5
- **Moderately/low-engaged users:** U2, U4

| User | Views | Clicks | Purchases | Behavior            |
|------|-------|--------|-----------|---------------------|
| U1   | 50    | 20     | 5         | Highly engaged      |
| U2   | 30    | 10     | 2         | Moderate engagement |
| U3   | 60    | 25     | 6         | Highly engaged      |
| U4   | 20    | 8      | 1         | Low engagement      |
| U5   | 55    | 22     | 5         | Highly engaged      |

### Step 3: Recommendation

The recommendation system can use cluster membership to identify users with similar interests and behaviors.

For example:

- Highly engaged users can receive **premium products and related-product recommendations**.
- Moderate users can receive **popular products and personalized offers**.
- Low-engagement users can receive **discounts and products related to their viewed items**.

### Benefits

Clustering improves recommendation systems by:

1. Finding users with similar behavior.
2. Providing personalized recommendations.
3. Increasing click-through and purchase rates.
4. Reducing irrelevant recommendations.
5. Supporting targeted promotions.

### Conclusion

**K-Means clustering** can effectively group users according to their online shopping behavior and help the e-commerce platform provide **personalized and relevant product**

recommendations.

### Q5. Traffic Data – Congestion Zone Detection

A clustering approach can identify locations with similar traffic conditions using **Vehicle Count, Average Speed, and Delay**.

#### Step 1: Normalize Features

The three variables have different scales, so they should be **standardized** before clustering.

#### Step 2: Apply K-Means

A suitable choice is **K = 2**, producing two major traffic segments.

| Location | Vehicles | Speed | Delay | Traffic Condition |
|----------|----------|-------|-------|-------------------|
| L1       | 200      | 30    | 10    | Low congestion    |
| L2       | 500      | 15    | 25    | High congestion   |
| L3       | 220      | 28    | 12    | Low congestion    |
| L4       | 600      | 10    | 30    | High congestion   |
| L5       | 210      | 32    | 9     | Low congestion    |

#### Step 3: Identify Clusters

##### Cluster 1 – Low Congestion:

L1, L3, L5

- Lower vehicle count
- Higher average speed
- Lower delay

##### Cluster 2 – High Congestion:

L2, L4

- High vehicle count
- Low average speed
- High delay

Therefore, **L2 and L4 are the main congestion zones**.

#### Step 4: Traffic Management

The traffic authority can use these clusters to:

- Adjust traffic signal timings.
- Divert vehicles to alternative routes.
- Increase traffic monitoring in congested areas.
- Improve road capacity where necessary.
- Provide real-time congestion alerts.
- Plan public transportation routes more effectively.

#### Conclusion

K-Means clustering can identify **high- and low-congestion zones** based on vehicle count, speed, and delay. In this dataset, **L2 and L4 form the high-congestion cluster**, helping traffic authorities take targeted actions to reduce congestion and improve traffic flow.